

# 2038 — Electrical and Electronics Practice Lab

Semester II

Credits: 1.5

Lab Course

Contact Hours: 45

## Official Source Declaration

This lesson is architected based on the official **SITTTR Kerala Revision 2026** syllabus for course code **2038 (Electrical and Electronics Practice Lab)**.

**Authoritative Source:** SITTTR Syllabus PDF (2038\_EE\_S2.pdf)

**Scope:** Covers all 4 Modules, 13 Experiments, and Open-ended experiments as prescribed.

## Course Dashboard

### Teaching Scheme

- **Practical:** 3 Hours/Week
- **Self-Learning:** 1.5 Hours/Week
- **Total Semester Hours:** 45

## Assessment Scheme

- **CIE (Internal):** 60 Marks
- **ESE (External):** 40 Marks
- **Total Marks:** 100

This course provides a hands-on foundation in electrical and electronic systems, covering safety, wiring, protective devices, electronic components, soldering, and Arduino programming.

## How to Study This Lesson

- 1 Safety First:** Before touching any equipment, read the safety guidelines in Module 2.
- 2 Theory Sync:** Correlate these experiments with Course 1031 (Basics of Electrical & Electronics Engineering).
- 3 Record Maintenance:** Tabulate all observations accurately as per the CIE evaluation criteria.

## Course Outcomes (CO)

| CO Code | Outcome Description                                                                | Bloom's Level |
|---------|------------------------------------------------------------------------------------|---------------|
| CO1     | Practice basic Electrical Measurements and Wiring Tasks using standard instruments | Apply         |
|         |                                                                                    |               |

| CO Code | Outcome Description                                                                                                | Bloom's Level |
|---------|--------------------------------------------------------------------------------------------------------------------|---------------|
| CO2     | Illustrate Electrical Safety Practices with the help of Protection Devices                                         | Apply         |
| CO3     | Practice Assembling, connection, testing, and verification of simple electronics circuits on a breadboard and PCB  | Apply         |
| CO4     | Demonstrate teamwork and problem solving skills in controlling simple electronic circuits with the help of Arduino | Apply         |

## Syllabus Coverage Map

| Module     | Focus Area                       | Experiments                              | Hours |
|------------|----------------------------------|------------------------------------------|-------|
| Module I   | Measurements & Home Installation | Multimeter, Extension Box, Fan, Inverter | 9     |
| Module II  | Domestic Wiring & Protection     | Distribution Box, Control Panels         | 9     |
| Module III | Electronics Assembly             | Soldering, Rectifiers, Clippers          | 9     |
| Module IV  | Arduino Programming              | LED Blinking, Voice Control, Open Ended  | 18    |

## Learning Roadmap

The course follows a logical progression from **Measurement** to **Installation**, then to **Electronic Hardware**, and finally **Software-controlled Hardware (Arduino)**.

Measurements

Wiring & Safety

Electronics/PCB

Arduino Control

Module I

# Electrical Measurements & Home Installations

Instructional Hours: 9 | CO Mapped: CO1

## Expt 1: Testing and Measuring Equipment

This experiment focuses on using the **Digital Multimeter (DMM)** and **Clamp Meter**.

- **Multimeter:** Measuring AC/DC Voltage, Resistance, and Continuity.
- **Non-Contact Voltage (NCV):** Detecting live wires without physical contact.
- **Clamp Meter:** Measuring high AC current by clamping around a single conductor (works on the principle of Mutual Induction).

### Malayalam support:

ഓൾട്ടിമീറ്റർ, നോൺ-കൺടാക്ട് വോൾട്ടേജ് (NCV) ടെസ്റ്റർ, ക്ലാമ്പ് മീറ്റർ എന്നിവ ഉപയോഗിച്ച് വൈദ്യുതി അളവുകൾ എടുക്കുന്നതിനും സുരക്ഷാ നടപടികൾ എടുക്കുന്നതിനും ഈ പരീക്ഷണം ശ്രദ്ധിക്കുന്നു. ക്ലാമ്പ് മീറ്റർ പ്രവർത്തിക്കുന്നത് പ്രതിരോധം അളക്കുന്നതിനുള്ള പ്രതീകാത്മകതയെ അടിസ്ഥാനമാക്കിയാണ്.

## Expt 2: Extension Box Wiring

Wiring two sockets controlled by separate switches. Key focus: **Wire sizing** and **Earthing**.

**Safety Check:** Always connect the Phase (Line) wire to the switch. The Neutral wire goes directly to the socket. The Earth wire (Green) must be connected to the larger top pin of the socket.

### Expt 3 & 4: Fan Connections & Inverters

**Ceiling Fan:** Wiring an induction motor fan with a regulator. Modern upgrade: Replacing with a **BLDC (Brushless DC) fan** which is more energy-efficient.

**Home Inverter:** Connecting a battery (Tall Tubular) to an inverter to provide backup for critical loads like lights and fans.

Module II

## Domestic Wiring & Protection

Instructional Hours: 9 | CO Mapped: CO2

### Expt 5: Distribution Box (DB) Wiring

Familiarization with the path of electricity: **Service Wire → Energy Meter → Main Switch/Isolator → RCCB → MCB → Sub-circuits**.

| Protection Devices |                           |                                             |
|--------------------|---------------------------|---------------------------------------------|
| Device             | Full Form                 | Function                                    |
| MCB                | Miniature Circuit Breaker | Protection against Overload & Short Circuit |

RCCB

Residual Current Circuit  
Breaker

Protection against Earth Leakage & Electric  
Shock

## Expt 6: Control Panels & Isolation

Demonstration of control panels to isolate high power circuits from low power control circuits for safety.

### Malayalam support:

റിക്കിഡുവെൻസ് റെസിഡുവൽ കറന്റ് സർക്യൂട്ട് ബ്രേക്കർ (RCCB) ഉപയോഗിക്കുന്നു. ഇത് ഹൈ പവർ സർക്യൂട്ടുകൾക്ക് ലോ പവർ കൺട്രോൾ സർക്യൂട്ടുകളിൽനിന്ന് വേർതിരിക്കാൻ സഹായിക്കുന്നു. MCB (Miniature Circuit Breaker) റെസിഡുവൽ കറന്റ് സർക്യൂട്ടുകൾക്ക് മാത്രമായി ഉപയോഗിക്കുന്നു.

Module III

# Electronics Assembly & Circuits

Instructional Hours: 9 | CO Mapped: CO3

## Expt 7: Breadboard & Soldering

**Breadboard:** Used for temporary circuit prototyping without soldering.

**PCB Soldering:** Permanent assembly. Practice de-soldering using a pump or wick.



PCB Tracks Representation

## Expt 8 & 9: Rectifiers & Clippers

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**Rectifiers:** Converting AC to DC.

- Half Wave: Uses 1 diode.
- Full Wave (Bridge): Uses 4 diodes.

**Clippers:** Circuits used to "clip" or remove portions of a waveform above or below a certain level.

Module IV

# Arduino Controller & Programming

Instructional Hours: 18 | CO Mapped: CO4

## Expt 10-13: Arduino Basics

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Introduction to **Arduino UNO** and the **Arduino IDE**.

- **LED Blinking:** Writing the first 'Hello World' of hardware.
- **Alternating LEDs:** Logic for flashing two LEDs.
- **Voice Control:** Interfacing Arduino with a voice module or smartphone to control lights.

### Malayalam support:

Arduino നോൺ പ്രോഗ്രാമിംഗ് സാധനം. നോൺ പ്രോഗ്രാമിംഗ് സാധനം പ്രോഗ്രാമിംഗ്, പ്രോഗ്രാമിംഗ് സാധനം പ്രോഗ്രാമിംഗ് സാധനം. IDE പ്രോഗ്രാമിംഗ് സാധനം പ്രോഗ്രാമിംഗ് സാധനം.

## Open Ended Experiment

Students must design one experiment independently. Suggested ideas from syllabus:

1. Hanging bed switch for lamp control.
2. Lamp test holder with fuse.
3. 0 to 5V variable DC supply.
4. Clap light using Arduino.

## Student Activities (Self-Learning)

| Week  | Activity Description                              | Hours |
|-------|---------------------------------------------------|-------|
| 1     | Chart of IEC vs IEEE electrical symbols           | 2     |
| 2     | Market survey: Cost of wires, switches, sockets   | 1.5   |
| 5     | Identify components in your home Distribution Box | 2     |
| 10    | Resistor color code examples (3, 4, 5, 6 band)    | 2     |
| 12-14 | DIY Arduino Programs (Student's choice)           | 4.5   |



# Formula & Procedure Bank

**Ohm's Law (Context: Resistor selection for LED)**

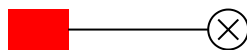
$$R = (V_{\text{supply}} - V_{\text{led}}) / I_{\text{led}}$$

Example: To run a 2V LED from 5V at 20mA:  $R = (5-2)/0.02 = 150\Omega$

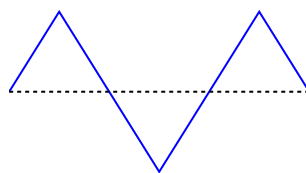
## Standard Wiring Colors (India)

- **Phase:** Red / Yellow / Blue
- **Neutral:** Black
- **Earth:** Green or Green-Yellow

## Visual Library



*Simple Switch-Lamp Circuit*



## Assessment Methodology

| Component | Method                                                             | Marks |
|-----------|--------------------------------------------------------------------|-------|
| CIE (60)  | Attendance (5), Continuous Eval (30), CA1 (15), CA2 (15), OEE (10) | 60    |
| ESE (40)  | End Semester Practical Exam                                        | 40    |

*Note: Continuous evaluation includes preparation, setup, observation, analysis, and viva.*

## Module-wise Q&A

What is the purpose of a Clamp Meter? (Module 1)

A clamp meter is used to measure current in a live conductor without breaking the circuit. It uses the principle of magnetic induction.

Difference between MCB and RCCB? (Module 2)

MCB protects against overcurrent and short circuits. RCCB protects against earth leakage current and prevents electric shocks to humans.

Why is a resistor used with an LED? (Module 3)

LEDs have low internal resistance. A series resistor is used to limit the current flowing through the LED to prevent it from burning out.

What are the two main functions in an Arduino sketch? (Module 4)

1. `setup()`: Runs once to initialize settings. 2. `loop()`: Runs repeatedly to control the logic.

## Practice Model Paper

*(Non-official practice guide)*

### Part A: Write-up (10 Marks)

Write the aim, circuit diagram, and procedure for constructing a Bridge Rectifier on a breadboard.

### Part B: Execution (10 Marks)

Wire up a single-phase ceiling fan with a regulator and demonstrate its speed control.

### Part C: Arduino (10 Marks)

Write a code to blink an LED with a 2-second delay and upload it to the board.

### Part D: Viva (10 Marks)

Explain the working of an RCCB.

## Quick Revision & Checklist

### Key Points

- Always check continuity before powering a circuit.
- Phase wire must always go through the switch.
- Use 1N4007 diodes for general rectification.

- Arduino pins can provide max 40mA (keep it safe).

## Exam Checklist

- [ ] Lab Record Certified?
  - [ ] Multimeter Probes checked?
  - [ ] Soldering Iron cleaned?
  - [ ] Arduino IDE installed?
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## Glossary

### Breadboard

A construction base for prototyping electronics.

### NCV

Non-Contact Voltage detection.

### BLDC

Brushless Direct Current motor.

### IDE

Integrated Development Environment (software for coding).

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## References

- *Fundamentals of Electrical and Electronics Engineering* - Susan S. Mathew, Saji T. Chacko.
  - *Basic Electrical Lab Manual* - S. K. Sahdev.
  - *Electronics Lab Manual* - K.A. Navas.
  - *Arduino made Simple* - Ashwin Pajankar.
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